

Magnus Victor Boock

📍 Odense Denmark

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Education

PhD	University of Southern Denmark , Computer Science	Aug 2025 – July 2028 (exp.)
	• Research area: Offline Distributional Reinforcement Learning with PAC-Bayesian methods. Supervised by Prof. Melih Kandemir. • Preprint: <i>Hitting Time Isomorphism for Multi-Stage Planning with Foundation Policies</i>. arXiv. A new theoretical framework and algorithm for long-horizon offline RL planning. Proposed and proved an operator-theoretic framework that learns asymmetric, triangle-inequality-satisfying state representations from hitting-time observations, then used it to develop Isomorphic Embedding Learning (IEL) — a foundation policy algorithm achieving state-of-the-art results on long-horizon navigation benchmarks via directed graph-based planning. • RLSS 2026: Selected attendee, Reinforcement Learning Summer School, Milan — competitive admission, lectures from leading RL researchers.	
MS	University of Southern Denmark , Computer Science	Sept 2023 – June 2025
	• Thesis: <i>Artificial data generation of Spatial Transcriptomics with Reinforcement Learning</i>. Generative Flow Networks for realistic tissue data generation. Supervised by Prof. Richard Röttger and Prof. Melih Kandemir. • Grades: Weighted average Danish grade: 10.67/12, GPA converted: 3.76/4. Thesis received the top grade.	
BS	University of Southern Denmark , Computer Science	Sept 2020 – June 2023
	• Thesis: Development of age estimation platform for skeletal remains (TA3). Supervised by Assoc. Prof. Luís Cruz-Filipe. • Grades: Weighted average Danish grade: 10.38/12, GPA converted: 3.59/4. Thesis received the top grade.	

Experience

ForeFlight , Routing Team Student Software Developer (.NET)	Odense, DK
• Enhanced routing engine with shortest path and constraint algorithms. • Implemented new functionality, bug fixes and ported internal tools.	March 2024 – April 2025
NIJ-TA3 ADBOU / SDU , Student Software Developer	Odense, DK
• Co-developed cross-platform desktop app for accurate age estimation based on skeletal remains. • The project is based on Transition Analysis 3, a collaborative effort between SDU & Univ. of Pennsylvania, funded by US National Institute of Justice. • The project has resulted in a publication in Forensic Science International.	July 2023 – 2024
Teaching Assistant at IMADA/SDU	Odense, DK
• Introduction to Computer Science, Algorithms and Data Structures, Networks and Cybersecurity and Reinforcement Learning	Sept 2021 – Present

Technologies

Languages: Python, C#, C++, C, JavaScript, TypeScript, Java, SQL

Frameworks & Tools: PyTorch, JAX, .NET, Git